

pattern is asymmetric, with regions of the embryo differing in both the number and size of newly formed cells.

Cleavage Pattern in Frogs

In the eggs of frogs (and many other animals), stored nutrients called **yolk** are concentrated toward one pole, called the **vegetal pole**, and away from the opposite or **animal pole**. This asymmetric distribution of yolk not only gives the two halves of the egg—the animal and vegetal hemispheres—different colors, but also influences the pattern of cleavage divisions, for reasons we will explore next.

When an animal cell divides, an indentation called a *cleavage furrow* forms in the cell surface as cytokinesis divides the cell in half (**Figure 47.7**). In the frog embryo, the first two cleavage furrows form parallel to the line (or meridian) connecting the two poles. During these divisions, the dense yolk slows completion of cytokinesis. As a result, the first cleavage furrow is still dividing the yolky cytoplasm in the vegetal hemisphere when the second cell division begins. Eventually, four blastomeres of equal size extend from the animal pole to the vegetal pole.

During the third division, the yolk begins to affect the relative size of cells produced in the two hemispheres. This division is equatorial (perpendicular to the line connecting the poles) and produces an eight-cell embryo. However, as each of the four blastomeres begins this division, yolk near the vegetal pole displaces the mitotic apparatus and the cleavage furrow from the egg equator toward the animal

pole. The result is a smaller blastomere size in the animal hemisphere than in the vegetal hemisphere. The displacing effect of the yolk persists in subsequent divisions, causing the blastocoel to form entirely in the animal hemisphere (see **Figure 47.7**).

Cleavage Patterns in Other Animals

Although yolk affects where division occurs in the eggs of frogs and other amphibians, the cleavage furrow still passes entirely through the egg. Cleavage in amphibian development is therefore said to be *holoblastic* (from the Greek *holos*, complete). Holoblastic cleavage is also seen in many other groups of animals, including echinoderms, mammals, and annelids. In those animals whose eggs contain relatively little yolk, the blastocoel forms centrally and the blastomeres are often of similar size, particularly during the first few divisions of cleavage (see **Figure 47.6**). This is the case for humans.

Yolk is most plentiful and has its most pronounced effect on cleavage in the eggs of birds and other reptiles, many fishes, and insects. In these animals, the volume of yolk is so great that cleavage furrows cannot pass through it, and only the region of the egg lacking yolk undergoes cleavage. This incomplete cleavage of a yolk-rich egg is said to be *meroblastic* (from the Greek *meros*, partial).

For chickens and other birds, the part of the egg that we commonly call the yolk is actually the entire egg cell. Cell divisions are limited to a small whitish area at the animal pole. These

divisions produce a cap of cells that sort into upper and lower layers. The cavity between these two layers is the avian version of the blastocoel.

In *Drosophila* and most other insects, yolk is found throughout the egg. Early in development, multiple rounds of mitosis occur without cytokinesis. In other words, no cell membranes form around the early nuclei. The first several hundred nuclei spread throughout the yolk and later migrate to the outer edge of the embryo. After several more rounds of mitosis, a plasma membrane forms around each nucleus, and the embryo, now the equivalent of a blastula, consists of a single layer of about 6,000 cells surrounding a mass of yolk (see **Figure 18.22**). Given that the number of cleavage divisions varies among animals, what determines the end of the cleavage stage? The **Scientific Skills Exercise** explores one of the landmark studies addressing this question.



▼ **Figure 47.7 Cleavage in a frog embryo.** The cleavage planes in the first and second divisions extend from the animal pole to the vegetal pole, but the third cleavage is perpendicular to the polar axis. In some species, the first division bisects the gray crescent, a lighter-colored region that appears opposite the site of sperm entry.

Eight-cell stage (viewed from the animal pole). The large amount of yolk displaces the third cleavage toward the animal pole, forming two tiers of cells. The four cells near the animal pole (closer, in this view) are smaller than the other four cells (colorized SEM).

Blastula (at least 128 cells). As cleavage continues, a fluid-filled cavity, the blastocoel, forms within the embryo. Because of unequal cell division, the blastocoel is located in the animal hemisphere. Both the drawing and the micrograph (assembled from fluorescence images) show cross sections of a blastula with about 4,000 cells.

